

Topics to be covered: Descriptive Statistical Analysis and Regression Models

1. Load [bank-full.csv](#) dataset and calculate the following descriptive statistics: -
 - a) Mean, standard deviation, mode, sum (make use of histograms to provide a visual analysis and calculate their values)
 - b) Covariance, Correlation between features (make use of scatter plots for visual analysis and calculate coefficients like Pearson correlation, and Spearman rank correlation)
 - c) Median, quartiles, max, min (make use of box plots for visual analysis and calculate their values)
2. On the bank-full.csv dataset, create a logistic regression model for predicting target variable (subscribed a term deposit or not). Use appropriate evaluation methods and report the performance of the model using appropriate metrics.
3. Load the [Energy Efficiency dataset](#), and split it into training and test datasets. Print the number of training and test samples in each of the datasets.
4. Predict Heating Load [Energy Efficiency dataset](#) with Linear Regression
 - 6.1 Train a linear regression model to predict Heating Load (Y1) using the original input features (X1–X8). Use the training set from Q5.
 - 6.3 Evaluate your model using the following metrics:
 - 6.3.1 Mean Absolute Error (MAE)
 - 6.3.2 Root Mean Squared Error (RMSE)
 - 6.6 Calculate the R^2 Score
 - 6.7 Please explain the following questions:
 - 6.7.1 Why did you choose these metrics?
 - 6.7.2 How well does your model perform based on these results?